



Fetomaternal hemorrhage: Pathogenesis, diagnosis, and clinical management

Deepika Sankaran¹, Evan Giusto^{1,2}, Michelle J. Lim³, Rebecca Valdez¹, Satyan Lakshminrusimha¹

¹Division of Neonatology, Department of Pediatrics, University of California, Davis, Sacramento, CA.

²Patient Care Services Advanced Practice: Neonatal, University of California Davis, Sacramento, CA., USA.

³Division of Pediatric Critical Care, Department of Pediatrics, University of California, Davis, Sacramento, CA.

Abstract

Clinically significant fetomaternal hemorrhage (FMH) can have devastating consequences on the newborn infant. Acute FMH close to childbirth can present with hypovolemia, shock, metabolic acidosis, and encephalopathy. Chronic FMH can be associated with congestive heart failure, pulmonary edema, hydrops and hepatomegaly. Early recognition and timely management of FMH are crucial in improving outcomes. This review article summarizes the epidemiology, pathogenesis, diagnosis, management and outcomes of clinically significant FMH. Current knowledge gaps in diagnosis and management of FMH are additionally described.

INTRODUCTION

Fetomaternal hemorrhage (FMH) is the loss of fetal blood into the maternal circulation during pregnancy or delivery. Under normal physiologic conditions, there is a small amount of fetal to maternal blood cell transfer across the placenta^{1,2}. Acute FMH refers to rapid fetal blood loss resulting in signs of fetal decompensation with abnormal fetal heart tracing (FHT). Moderate-to-severe FMH can result in significant fetal compromise due to the combination of fetal anemia, hypovolemia, metabolic acidosis, and hypoxia culminating in emergent delivery. FMH can often occur without an obvious precipitating event. Acute, severe FMH may necessitate extensive neonatal resuscitation, often with impaired or delayed response to routine resuscitative measures. Emergent blood transfusion may be necessary, as FMH can result in severe neonatal anemia, hypoxic ischemic encephalopathy,

Address correspondence to: Deepika Sankaran, MD FAAP. Department of Pediatrics, 2516 Stockton Blvd, Sacramento, CA 95817. Fax: 916-456-2236. dsankaran@health.ucdavis.edu.

Author Contribution statement: DS conceptualized, designed, drafted the initial manuscript, reviewed and revised the manuscript. EG, ML, and SL contributed to the concept, reviewed and revised the manuscript. All the authors have critically revised and approved the final version of the manuscript. All authors agree to be accountable for all aspects of the work.

Conflict of interest: The funders had no role in the design of the study; in the collection, analyses, or interpretation of data; in the writing of the manuscript, or in the decision to publish the results.

Conflict of Interest: The authors have no conflicts of interest to disclose.

and poor long-term neurodevelopmental outcomes. Severe FMH if untreated can result in intrauterine fetal death or still birth.^{2,3} In contrast, the clinical presentation of subacute or chronic FMH (occurring over several hours to days to weeks) may be indolent due to fetal adaptative changes to lower circulating blood volume but can present with anemia, congestive cardiac failure and hydrops in severe cases.⁴

EPIDEMIOLOGY

The incidence of FMH is difficult to determine due to variability in clinical definitions, differences in diagnostic methods, and occult presentations often leading to missed or delayed diagnoses. There is wide variation in the reported incidence depending on the volume of estimated blood loss, ranging from 3 per 1000 live births^{5–7} for ≥ 30 mL blood loss to as infrequent as 0.1 per 1000 live births for larger volumes of fetal blood loss.^{8,9} Standardizing the reporting of clinically significant FMH by combining criteria such as estimated fetal blood volume loss (either per kilogram body weight or percentage of fetal blood volume), rapidity of blood loss, and presence of abnormal fetal heart tracing or neonatal decompensation requiring extensive resuscitation including use of volume expanders are needed to study incidence and outcomes. Further, up to 40% cases of FMH can be missed.¹⁰ In a recent registry-based study, Halling et al reported use of volume bolus in 64/233 (27.5%) of newborns requiring chest compressions, compared to 561/1153 (49%) requiring epinephrine.¹¹ Early recognition of FMH and administration of volume expanders may hasten return of spontaneous circulation (ROSC) and restore cerebral and coronary blood flow during resuscitation potentially improving neonatal outcomes.

Stillbirth, neonatal anemia, and cesarean section are associated with FMH but are likely to be consequences of FMH rather than contributing factors to its occurrence. Obstetrical events associated with FMH include placenta previa, fetal trauma, monochorionic monoamniotic twins,² umbilical cord prolapse, tight nuchal cord,¹² external cephalic version,¹³ maternal abdominal trauma,¹⁴ placental abruption, preeclampsia, placental tumors,¹⁵ and amniocentesis (Table 1).^{16–25} While placental abruption and FMH can occur concurrently, placental abruption is not more commonly observed in cases of FMH.²⁶ Moreover, FMH can occur in pregnancies without any of the above associated factors.²

PATHOPHYSIOLOGY OF FETOMATERNAL HEMORRHAGE

Human placenta is hemochorial with maternal uterine spiral arteries pouring oxygenated blood into the intervillous space where fetal villi (with fetal vessels) “bathe” leading to gas exchange (figure 1). The precise mechanism of FMH is unclear. A histological study of placentas demonstrated a higher frequency of parenchymal and retroplacental bleeds, intervillous thrombi and infarction that may increase the likelihood of fetal cells crossing over to the maternal circulation. The severity of these abnormalities appeared to correlate with the volume of FMH.²⁷ It is likely that fetal blood loss into the intervillous space can lead to absorption into maternal circulation in the presence of placental pathology such as trauma, infarction, or thrombosis (figure 1). Exposure to fetal cells may result in maternal sensitization to fetal antigens such as Rh, and appropriate therapy with Rh immunoglobulin may be needed.

The two potential pathophysiologic mechanisms of FMH are as follows:

- a. Acute FMH (figure 1): Acute loss of fetal blood into maternal circulation can cause rapid clinical deterioration due to acute hypovolemia, decrease in preload and end-diastolic ventricular volume. Cardiac output is the product of stroke volume and heart rate, and stroke volume is dependent on left ventricular preload (end-diastolic ventricular volume), cardiac contractility and afterload (or systemic vascular resistance). Asphyxiated myocardium in a newborn is not able to increase intrinsic contractility owing to depletion of energy reserves. The newborn heart is presumed to function in the flat (or plateau) portion of the Frank Starling curve, and heart rate is thought to be the predominant determinant of the cardiac output rather than stroke volume.²⁸ In a perinatal term ovine model of hemorrhagic asphyxia, there was dramatic decrease in both stroke volume and carotid arterial blood flow during acute fetal blood loss of 45 mL/kg over 5 minutes suggesting that some fetal hearts may function in the steep part of the Frank Starling's law although this speculation may require further investigation in other models.²⁹ Interestingly, the fetal heart rate was minimally affected during acute hemorrhage from the ovine fetus (figure 2a) suggesting that FHT may not be an early sign of fetal blood loss.²⁹ In term lambs with invasive monitoring of heart rate, mean blood pressure and left carotid artery blood flow, removal of 45-mL/kg of blood from the fetoplacental circulation was compared to control lambs without blood loss.²⁹ Considering 100–120-mL/kg as the total fetoplacental blood volume, 45 mL/kg represents about a third to half of the fetoplacental blood volume that is lost acutely over 5 minutes. During the period of blood loss, the mean left carotid artery blood flow decreased significantly, hastening cardiac arrest following umbilical cord occlusion when compared to lambs who had asphyxia without blood loss.²⁹ Notably, the heart rate did not change significantly during the period of blood loss (figure 2a), despite the removal of 45-mL/kg of blood that significantly decreased mean blood pressure, cerebral regional oxygen saturation, carotid artery blood flow and stroke volume (figures 2b–2e). Therefore, fetal heart rate monitoring may have limitations in predicting fetal blood loss.²⁹

Tight nuchal cord is a unique situation where acute fetal blood loss may occur. Tight nuchal cord is defined as 360° wrapping of umbilical cord around the fetal neck³⁰ and by the relative inability to manually reduce the loop over the fetal head. It is reported in about 6.6% of live births.^{29,30} The thin-walled collapsible umbilical vein can be compressed relatively easily by tight nuchal cord compared to the thick-walled umbilical arteries (figure 3).²⁹ The fetus is compromised by asphyxia due to decrease umbilical venous blood flow. The thick-walled umbilical arteries may remain patent with potential loss of fetal blood into the placenta (fetoplacental hemorrhage). This may result in hypovolemia, anemia and loss of left ventricular preload.^{29,31} Further studies are warranted to evaluate the fetal vs. placental volume changes and pathophysiology of tight nuchal cord and other cord accidents. Although a tight nuchal cord and cord rupture may not lead to FMH, clinical presentation can be similar due to fetal blood loss.

- b. Subacute or chronic FMH (figure 4): Fetal blood loss could be subacute if it occurs over hours to days and chronic if it occurs over days to weeks. With slower blood loss as seen in subacute and chronic FMH, the fetus may adapt to lower circulating blood volume by a rise in heart rate to maintain adequate cardiac output, and by equilibration from reverse flow of blood from the placenta into the fetus and increased red blood cell production.⁴ The fetoplacental circulation consists of ~120–135 mL/kg³² which is approximately 1.5-times the fetal circulating blood volume, allowing compensation for the fetal hypovolemia. The newborn infant may present with anemia, with clinical signs and symptoms of heart failure, and in severe cases with hydrops. Thus, anemia should be corrected slowly with judicious blood transfusions with diuretic use or reverse partial exchange (removal of neonatal blood and replacement with packed red blood cells – PRBCs in small aliquots)³³ in neonates with subacute and chronic FMH to avoid worsening of pulmonary edema and congestive cardiac failure.³⁴ Severe chronic fetal anemia can result in hydrops fetalis. Laboratory evaluation shows features of chronic anemia with reticulocytosis.

INTRAPARTUM RECOGNITION OF FETOMATERNAL HEMORRHAGE

Pregnant people may seek emergent obstetrical care due to decreased or absent fetal movements or may present in labor with subsequent diagnosis of FMH from routine fetal monitoring.⁴ Monitoring of FHT is the standard of care to assess fetal wellbeing.³⁵ Abnormalities in FHT may allow early recognition of clinically significant FMH, although normal FHT does not rule out FMH.² Initial FHT changes in FMH can be non-specific and include loss of beat-to-beat variability or tachycardia. Sinusoidal pattern of FHT characterized by visually apparent, smooth, sine-like undulating pattern in FHR baseline with a cycle frequency of three to five cycles per minute that persists for ≥ 20 minutes, is pathognomonic of significant FMH resulting in fetal anemia and hypoxia.³⁵ However, sinusoidal FHT may be observed only in severe cases.³⁶ Vigilance should be exercised for sinusoidal pattern that is intermittent (or non-persistent) as it may be an early ominous sign of fetal anemia.³⁷ Category III FHT (absence of normal FHR variability and presence of either recurrent late or variable decelerations or fetal bradycardia) may also be observed in FMH, and is associated with high likelihood of severe hypoxia and metabolic acidosis and would indicate emergent delivery.³⁷ In addition to FHT, elevated peak systolic velocity and reversed end-diastolic flow in the middle cerebral artery on doppler study of the fetus has been used to detect fetal anemia but has low specificity with a false positivity rate of 60%.³⁸ Fetal growth restriction and rarely hydrops may occur with chronic FMH.²

Umbilical artery dopplers with increased blood flow velocity in the umbilical vein and normal umbilical artery pulsatility indices have been reported in massive FMH.³⁹ Increase in detection of nucleated fetal red blood cells may be observed in response to subacute or chronic FMH by means of non-invasive pregnancy tests (NIPT). Increase in maternal serum α -fetoprotein has been used in combined with flow cytometry or Kleihauer-Betke (KB) test in the detection of FMH.¹⁶

CLINICAL PRESENTATION IN NEONATES AFTER FETAL BLOOD LOSS

In deliveries complicated by acute FMH, fetal trauma, umbilical cord prolapse, or tight nuchal cord, the newborn might present in a state of acute circulatory failure with poor tissue perfusion, hypovolemic shock and metabolic acidosis.⁴⁰ In cases of occult FMH without any obvious history of FMH or abnormality on fetal heart rate monitoring, clinical signs and symptoms of poor perfusion may be the only evidence of antecedent FMH. Newborns affected by acute FMH may present with persistently low heart rate or cardiac arrest, pallor, weak peripheral pulses, and delayed capillary refill time.⁴⁰ Conversely, the newborn may have tachycardia if FMH is recognized in a timely manner followed by prompt delivery. In a perinatal lamb model, acute removal of 20-mL/kg of fetoplacental blood volume from fetal lambs did not affect the heart rate although the carotid artery blood flow and mean blood pressure decreased during the period of blood loss (Fig. 5). Clinical assessments of perfusion may be suboptimal in differentiating neonates affected by acute FMH from asphyxiated infants without FMH or with subacute or chronic FMH. Furthermore, clinical evaluation of color of mucus membranes, capillary refill time, inspection of color and palpation of peripheral pulses are all subjective, not reproducible, and not validated and therefore unreliable. The newborn may have low hemoglobin and high base deficit in the first postnatal blood gas. The base deficit in neonatal arterial blood increased significantly with decreasing neonatal hemoglobin.³ Metabolic acidosis may be observed both in the umbilical arterial blood gas and in the newborn's first postnatal gas as a result of anerobic metabolism secondary to ischemia and hypoxia in the fetus. Lack of response to initial steps of resuscitation conducted as per the American Academy of Pediatrics Neonatal Resuscitation Program (AAP-NRP) guidelines (i.e., effective ventilation, chest compressions and intravenous epinephrine) may suggest hypovolemic shock and a trial of volume replacement may be considered.⁴⁰ Clinicians should be judicious with the use of volume expanders in euvoletic newborns and neonates with chronic FMH, as volume overload to an asphyxiated heart may worsen myocardial function and decrease cardiac output.

DIAGNOSIS OF FETOMATERNAL HEMORRHAGE

Flow cytometry is utilized in diagnosing and quantifying fetomaternal hemorrhage in hospitals in the United States. In flow cytometry, fetal hemoglobin is tagged with fluorescent antibody and the volume of FMH is quantified by an automated process, making it less tedious and more objective compared to the KB test, with quick results within 30 minutes.⁴ However, the equipment is expensive and may not be readily available in labor and delivery suites, thus delaying the time to obtain results. Flow cytometry also helps determine the appropriate dosage of Rh immunoglobulin for RhD-negative pregnancies to prevent RhD alloimmunization that can result in hemolytic disease of the newborn.⁴¹

The KB acid elution test is an alternate method used to detect fetal RBCs in the maternal blood sample. It is based on the principle that fetal RBCs contain predominantly fetal hemoglobin (HbF) which is resistant to acid elution in contrast to adult hemoglobin which is acid sensitive.⁴² A peripheral smear prepared from maternal blood sample is treated with an acid buffer followed by staining with hematoxylin, that results in pink-stained fetal RBCs surrounded by pale "ghost" outlines of maternal RBCs under microscopy (figure 1 inset).

This laborious technique involves manual counting of 10,000 RBCs followed by assessment of the percentage of fetal cells.⁴ Volume of FMH (in mL) is calculated (volume of FMH = (percentage of fetal cells determined by KB test/100) × 5,000 mL).⁴² The KB test is inexpensive (but labor intensive) and can be performed at bedside. However, KB test is subjective, error-prone, not reproducible, and cannot be standardized. Additionally, these techniques do not aid in assessing the rapidity and timing of FMH.

Hemoglobin measured from the umbilical cord arterial blood sample after delivery may help in early detection fetal anemia early after delivery. However, if hemorrhage is acute and occurred immediately prior to umbilical cord sampling, hemoglobin levels may be low-normal as there has been no time for equilibration. Placental pathology may provide additional supportive evidence but the results may be delayed. Assessing the weight of placenta after delivery to assess residual placental blood volume and comparing it to standard weight of placenta to determine the extent of FMH will need to be investigated.⁴³

MANAGEMENT OF NEONATES AFFECTED BY FMH AT DELIVERY

Deferred cord clamping (DCC) may need to be avoided or individualized in circumstances where the placental circulation is not intact, such as placental abruption, placenta previa, and umbilical cord avulsion. In neonates with intact placental circulation, and not in need of emergency resuscitation, may potentially benefit from DCC by receiving additional placental transfusion.²⁹ The benefit/risk of umbilical cord milking in term infants with FMH is not known. It may be beneficial in some cases of acute FMH if suspected before birth.

The current AAP-NRP guidelines recommend positive pressure ventilation and chest compressions followed by intravenous epinephrine for an asphyxiated neonate presenting with bradycardia with or without suspicion of FMH.⁴⁰ When there is persistently low heart rate not responding to effective ventilation, chest compressions and epinephrine, and clinical signs of hypovolemic shock or obvious history of clinically significant FMH, then a trial of volume expander (10ml/kg) over 5 to 10 minutes is recommended⁴⁰. Current guidelines are based on expert opinion (level of evidence C-EO (expert opinion), “moderate” or 2b class of recommendation, or benefits outweigh risks).⁴⁴ Although this approach of chest compressions followed by epinephrine is effective in normovolemic asphyxia, it is not known if it will be sufficient to allow adequate blood flow to the heart and brain during resuscitation in hypovolemic asphyxia. It is possible that chest compressions and epinephrine may be less effective in increasing the diastolic blood pressure (the primary driver of coronary perfusion) in hypovolemic asphyxia due to lack of adequate preload. Administration of a volume expander early during cardiopulmonary resuscitation may swiftly replenish the left ventricular end-diastolic volume, enhancing the effectiveness of chest compressions and intravenous epinephrine to possibly hasten return of spontaneous circulation. Prospective randomized clinical trials evaluating the timing, frequency, volume, and efficacy of volume expanders during neonatal resuscitation are challenging to perform due to the emergent clinical circumstances as well as ethical concerns.

Three factors should be taken into consideration before administering a fluid bolus during advanced resuscitation of a depressed infant with suspected FMH. How likely is asphyxia

secondary to hypovolemia? Is blood loss (such as FMH) likely to be acute, subacute or chronic? What is the gestational age of the neonate? There is paucity of evidence to support routine administration of volume expanders in a neonate without obvious FMH. Asphyxia that is refractory to ventilation, chest compressions, and epinephrine may be hypovolemic or cardiogenic. However, as FMH may be occult, a trial of volume replacement may be considered in those infants who do not respond to resuscitation with due caution, given the likelihood of worsening cardiac output due to volume overload on an asphyxiated heart and risk of pulmonary edema. In bradycardic preterm infants not responding to effective ventilation, chest compressions, and intravenous epinephrine, the role of volume expansion in the delivery room is not known, especially as they are at increased risk of intraventricular hemorrhage from fluctuations in cerebral blood flow due to their immature germinal matrix.

Transfusion using fresh O-negative blood products help restore circulating blood volume and oxygen carrying capacity following clinically significant FMH. Crystalloids (normal (0.9%) saline or lactated Ringer's solution) or fresh blood are preferred over albumin as the choice of volume expander in the delivery room.^{29,45} In piglets who had asphyxia, hemorrhage and cardiac arrest, use of normal saline was compared to administration of animal's own anticoagulated blood at 10-mL/kg over 2 minutes, with a maximum of 3 boluses. Interestingly, return of spontaneous circulation occurred before volume infusion in a quarter of the piglets, and there was no difference in the time to achieve return of spontaneous circulation between use of crystalloid and blood transfusion. In prospective studies from <24 hours old newborns in the neonatal intensive care unit comparing normal saline and 5% albumin infusion to manage systemic hypotension, normal saline was found to be equally effective, less expensive, readily available, and there was no evidence to favor the use of albumin over crystalloids. Among crystalloids, normal saline and lactated Ringer's solution are often readily available to use as volume expanders in the delivery room. In an adult swine model of hemorrhagic shock, use of normal saline as the volume expander resulted in greater volume of fluid administered, hyperchloremic metabolic acidosis and dilutional coagulopathy, when compared to lactated Ringer's solution.⁴⁶ In a model of moderate hemorrhage in rats, use of normal saline was associated with more metabolic acidosis, physiological derangement, and worse survival compared to use of lactated Ringer's solution.⁴⁷ Lactated Ringer's solution has lower chloride (109 mEq/L vs 154 mEq/L in normal saline) and has a more neutral pH (6.5 vs 5.0 in normal saline). Physiologically, lactated Ringer's solution may be a better crystalloid compared to normal saline for use as a volume expander in the delivery room. However, normal saline is often preferred over lactated Ringer's solution due to the risk of precipitation with blood products (owing to presence of calcium in lactated Ringer's solution) that may be needed during resuscitation. Newborns may have severe metabolic acidosis after hemorrhagic hypovolemic shock requiring management with fluid boluses to correct hypoperfusion to tissues. On the contrary, metabolic acidosis in the setting of euvolemia does not warrant volume expansion. Use of sodium bicarbonate is not recommended currently during neonatal resuscitation. However, after replenishing fluid losses, a small dose of slowly administered sodium bicarbonate at 1 mEq/kg may be considered for persistent and refractory metabolic acidosis in the neonatal intensive care unit (NICU). A recent cross-over trial study that compared

normal saline and lactated ringer's solution among hospitalized patients except newborns did not find any in rates of death or readmission to the hospital.⁴⁸

In subacute/ chronic FMH, intrauterine transfusions may be required for severe anemia in the fetus similar to the approach in anemia due to alloimmunization during pregnancy due to destruction of fetal RBCs by maternal antibodies.⁴⁹ Management of chronic FMH and anemia is based on severity of anemia and associated clinical features such as cardiac failure and hydrops (figure 4). Presence of effusions may necessitate emergency abdominal paracentesis, thoracocentesis or pericardiocentesis.^{50,51} Fluid replacement should be gradual to prevent worsening of pulmonary edema, and a partial exchange transfusion may be needed. In Rh-negative pregnancies, an appropriate dose of Anti-Rh immunoglobulin should be administered if the fetus/neonate is Rh-positive.

OUTCOMES IN NEONATES AFFECTED BY FMH

In a retrospective study of 48 fetuses with massive FMH (defined as ≥ 40 RBCs per 10,000 maternal RBCs) by Rubod et al, the incidence of fetal death was 6 (12.5%). Nine out of 42 liveborn infants required neonatal intensive care (21%). FMH of ≥ 20 mL/kg increased risk of fetal death, preterm delivery, transfer to neonatal intensive care unit, and neonatal anemia requiring transfusion. However, among survivors, there was no significant risk of neurological sequelae with only one infant showing signs of global hypotonia at discharge (which eventually turned out to be mitochondrial cytopathy).⁵² None of the other survivors available for follow-up had neurological sequelae. In the retrospective study by Kadooka et al study, lower neonatal hemoglobin (3.6 ± 1.4 vs. 5.4 ± 1.1 g/dL) and postnatal pH (7.09 ± 0.11 vs. 7.25 ± 0.13), and higher base deficits (17.5 ± 5.4 vs. 10.4 ± 6.0 mmol/L) were reported among infants with poor neurological outcome compared to infants with normal neurological outcome.³ In another retrospective study conducted in Japan over a 15-year period, 9/18 (50%) newborns with severe neonatal anemia and KB stain with $> 4\%$ fetal RBCs had poor neurological outcome defined as either cerebral palsy, mental retardation, attention deficit hyperactivity disorder or epilepsy at 12 months after birth.³

Most severe cases might culminate in unexplained intrauterine fetal death, miscarriage, and fetal hydrops.⁴ Lower initial neonatal hemoglobin is associated with poor outcomes (worse with hemoglobin < 3 g/dL vs < 5 g/dL and < 7 g/dL) including intraventricular hemorrhage, hypoxic ischemic encephalopathy, and death.⁹ Initial hemoglobin of < 5 g/dL after FMH is associated with need for resuscitation at birth, emergent blood transfusion, and risk of the above-mentioned morbidities and mortality.⁶

GAPS IN KNOWLEDGE ON FMH AT DELIVERY (Table 2)

Early and accurate recognition of fetal blood loss (hypovolemia and anemia) remains a knowledge gap (Table). Currently available tools are inadequate for early recognition of clinically significant FMH at time of delivery. It is also not possible to assess the rapidity of blood loss in FMH. Monitoring of fetal oxygenation status may have a role in early detection of FMH in addition to FHR monitoring and warrants further investigation. In cases with

obvious history of FMH, the optimal choice of volume expander, timing, volume and speed of administration need to be investigated (Table 2).

CONCLUSION

Clinically significant FMH may necessitate extensive neonatal resuscitation including use of volume expanders in the delivery room. In newborn infants with obvious clinical evidence of FMH not responding to ventilation, chest compressions and intravenous epinephrine, volume expanders are indicated in the delivery room. Early recognition of acute FMH is critical in determining the need for volume expanders in the delivery room whereas volume expanders should be avoided in normovolemic asphyxia and should be used cautiously in chronic FMH. Careful assessment of the infant during the post-resuscitation phase for need for volume is important. Following admission in the NICU, the incidence of systemic hypotension and potential need for vasopressors needs further investigation. Systemic hypotension in the setting of hypovolemia and anemia secondary to FMH, the use of norepinephrine or vasopressin may increase cardiac afterload and potentially increase the strain on the heart. On the contrary, use of lower doses of epinephrine or dopamine may increase blood pressure and systemic perfusion without increasing the afterload significantly and warrants future research.

Acknowledgements:

The authors would like to thank the funding sources listed below.

Financial disclosure:

The authors have no conflict of interest to declare. This work has been supported by Research Grant from the American Academy of Pediatrics- Neonatal Resuscitation program (AAP-NRP) (D.S., and E.G.), Eunice Kennedy Shriver National Institute of Child Health and Human Development (NICHD) and National Institutes of Health HD109443 (D.S.) and HD072929 (S.L.), National Heart, Lung and Blood Institute (NHLBI) 1K08HL181183-01 (D.S.), Children's Miracle Network (D.S.), and ZOLL Foundation Inc. (D.S. and E.G.). The project described was supported by the National Center for Advancing Translational Sciences, National Institutes of Health, through grant number UL1 TR001860 and linked award 5KL2TR001859 (D.S.) and 5KL2TR001859 (M.L.). The content is solely the responsibility of the authors and does not necessarily represent the official views of the NIH or AAP-NRP. The funder/ sponsor did not participate in planning or devising this work.

Funding information:

The authors have no conflict of interest to declare. This work has been supported by Research Grant from the American Academy of Pediatrics- Neonatal Resuscitation program (AAP-NRP) (D.S., and E.G.), Eunice Kennedy Shriver National Institute of Child Health and Human Development (NICHD) and National Institutes of Health HD109443 (D.S.) and HD072929 (S.L.), Children's Miracle Network (D.S.), and ZOLL Foundation Inc. (D.S. and E.G.). The project described was supported by the National Center for Advancing Translational Sciences, National Institutes of Health, through grant number UL1 TR001860 and linked award 5KL2TR001859 (D.S.), 1K08HL181183-01 (DS), and 5KL2TR001859 (M.L.). The content is solely the responsibility of the authors and does not necessarily represent the official views of the NIH or AAP-NRP. The funder/ sponsor did not participate in planning or devising this work.

REFERENCES

1. Bianchi DW Fetomaternal cell trafficking: a story that begins with prenatal diagnosis and may end with stem cell therapy. *Journal of pediatric surgery* 42, 12–18 (2007). [PubMed: 17208534]
2. Wylie BJ & D'Alton ME Fetomaternal hemorrhage. *Obstetrics & Gynecology* 115, 1039–1051 (2010). [PubMed: 20410781]

3. Kadooka M, Kato H, Kato A, Ibara S, Minakami H & Maruyama Y Effect of neonatal hemoglobin concentration on long-term outcome of infants affected by fetomaternal hemorrhage. *Early human development* 90, 431–434 (2014). [PubMed: 24964226]
4. Maier JT, Schalinski E, Schneider W, Gottschalk U & Hellmeyer L Fetomaternal hemorrhage (FMH), an update: review of literature and an illustrative case. *Archives of gynecology and obstetrics* 292, 595–602 (2015). [PubMed: 25778871]
5. Sebring E & Polesky H Fetomaternal hemorrhage: incidence, risk factors, time of occurrence, and clinical effects. *Transfusion* 30, 344–357 (1990). [PubMed: 2190367]
6. Sebring ES & Polesky HF Fetomaternal hemorrhage: incidence, risk factors, time of occurrence, and clinical effects. *Transfusion* 30, 344–357, doi:10.1046/j.1537-2995.1990.30490273444.x (1990). [PubMed: 2190367]
7. Rubod C, Deruelle P, Le Goueff F, Tunez V, Fournier M & Subtil D Long-term prognosis for infants after massive fetomaternal hemorrhage. *Obstet Gynecol* 110, 256–260, doi:10.1097/01.Aog.0000271212.66040.70 (2007). [PubMed: 17666598]
8. de ALMEIDA V & BOWMAN JM Massive fetomaternal hemorrhage: Manitoba experience. *Obstetrics & Gynecology* 83, 323–328 (1994). [PubMed: 8127519]
9. Christensen RD, Lambert DK, Baer VL, Richards DS, Bennett ST, Ilstrup SJ et al. Severe neonatal anemia from fetomaternal hemorrhage: report from a multihospital health-care system. *J Perinatol* 33, 429–434, doi:10.1038/jp.2012.142 (2013). [PubMed: 23196720]
10. Carr NR, Henry E, Bahr TM, Ohls RK, Page JM, Ilstrup SJ et al. Fetomaternal hemorrhage: Evidence from a multihospital healthcare system that up to 40% of severe cases are missed. *Transfusion* 62, 60–70, doi:10.1111/trf.16710 (2022). [PubMed: 34674275]
11. Halling C, Raymond T, Brown LS, Ades A, Foglia EE, Allen E et al. Neonatal delivery room CPR: An analysis of the Get with the Guidelines[®]-Resuscitation Registry. *Resuscitation* 158, 236–242, doi:10.1016/j.resuscitation.2020.10.007 (2021). [PubMed: 33080368]
12. Dunn PM Tight nuchal cord and neonatal hypovolaemic shock. *Arch Dis Child* 63, 570–571, doi:10.1136/adc.63.5.570-a (1988).
13. Boucher M, Marquette GP, Varin J, Champagne J & Bujold E Fetomaternal hemorrhage during external cephalic version. *Obstet Gynecol* 112, 79–84, doi:10.1097/AOG.0b013e318179978c (2008). [PubMed: 18591311]
14. Goodwin TM & Breen MT Pregnancy outcome and fetomaternal hemorrhage after noncatastrophic trauma. *American Journal of Obstetrics and Gynecology* 162, 665–671, doi:10.1016/0002-9378(90)90979-H (1990). [PubMed: 2316567]
15. Henningsen A-KA, Maroun LL, Havsteen H & Svare J Massive Fetomaternal Hemorrhage Caused by an Intraplacental Choriocarcinoma: A Case Report. *Case Reports in Medicine* 2010, 767218, doi:10.1155/2010/767218 (2010). [PubMed: 20204132]
16. Samura O, Mihar N, Hyodo M, Honda H, Ohashi Y, Honda N et al. Cell-free Fetal DNA in Maternal Circulation after Amniocentesis. *Clinical Chemistry* 49, 1193–1195, doi:10.1373/49.7.1193 (2003). [PubMed: 12816922]
17. Sikovanyecz J, Horváth E, Pásztor N, Keresztúri A, Szabó J & Pál A Fetomaternal transfusion after amniocentesis and cordocentesis. *Irish Journal of Medical Science* 180, 697–701, doi:10.1007/s11845-010-0663-3 (2011). [PubMed: 21190089]
18. Brandt JS & Ananth CV Placental abruption at near-term and term gestations: pathophysiology, epidemiology, diagnosis, and management. *American Journal of Obstetrics & Gynecology* 228, S1313–S1329, doi:10.1016/j.ajog.2022.06.059 (2023). [PubMed: 37164498]
19. Faiz AS & Ananth CV Etiology and risk factors for placenta previa: an overview and meta-analysis of observational studies. *J Matern Fetal Neonatal Med* 13, 175–190, doi:10.1080/jmf.13.3.175.190 (2003). [PubMed: 12820840]
20. Rincon M, Rhodes H, Boniface E, Pereira L, D’Mello RJ, Hayer S et al. The natural history and management of vasa previa: a single institution’s 15-year experience managing patients remote from labor and delivery. *BMC Pregnancy Childbirth* 25, 581, doi:10.1186/s12884-025-07708-8 (2025). [PubMed: 40382549]

21. Mendez-Figueroa H, Dahlke JD, Vrees RA & Rouse DJ Trauma in pregnancy: an updated systematic review. *American Journal of Obstetrics & Gynecology* 209, 1–10, doi:10.1016/j.ajog.2013.01.021 (2013). [PubMed: 23333541]
22. Duzyj CM, Einerson BD, Afshar Y, Fox KA, Saad AF, Carusi DA et al. Placenta Accreta Spectrum, Present and Future: The Eunice Kennedy Shriver National Institute of Child Health and Human Development Workshop Summary. *Obstet Gynecol* 145, 581–585, doi:10.1097/aog.0000000000005929 (2025). [PubMed: 40311146]
23. Kawakita T, Huang CC & Landy HJ Risk Factors for Umbilical Cord Prolapse at the Time of Artificial Rupture of Membranes. *AJP Rep* 8, e89–e94, doi:10.1055/s-0038-1649486 (2018). [PubMed: 29755833]
24. Chen R & Lin L Spontaneous Umbilical Cord Vascular Rupture during Labor: A Retrospective Analysis of 12 Cases. *Am J Perinatol* 42, 586–592, doi:10.1055/a-2412-3169 (2025). [PubMed: 39353592]
25. Silva GV, Gontijo CT, Lunguinho APC, Caetano MSG, Callado GY, Araujo Júnior E et al. Perinatal Outcomes Related to the Presence of a Nuchal Cord During Delivery: A Retrospective Cohort Study. *Diagnostics (Basel)* 15, doi:10.3390/diagnostics15101197 (2025).
26. Boller M, Navarrete E, Hung Y-Y, Weintraub MR, Moore GS & Schauer GM Placental Abruption Is Not Associated With Clinically Significant Fetomaternal Hemorrhage [290]. *Obstetrics & Gynecology* 135 (2020).
27. Devi B, Jennison RF & Langley FA Significance of placental pathology in transplacental haemorrhage. *Journal of Clinical Pathology* 21, 322–331, doi:10.1136/jcp.21.3.322 (1968). [PubMed: 4972435]
28. Vrancken SL, van Heijst AF & de Boode WP Neonatal Hemodynamics: From Developmental Physiology to Comprehensive Monitoring. *Front Pediatr* 6, 87, doi:10.3389/fped.2018.00087 (2018). [PubMed: 29675404]
29. Sankaran D, Lane ECA, Valdez R, Lesneski AL & Lakshminrusimha S Role of Volume Replacement during Neonatal Resuscitation in the Delivery Room. *Children (Basel)* 9, doi:10.3390/children9101484 (2022).
30. Henry E, Andres RL & Christensen RD Neonatal outcomes following a tight nuchal cord. *Journal of Perinatology* 33, 231–234, doi:10.1038/jp.2012.79 (2013). [PubMed: 22699359]
31. Peesay M Nuchal cord and its implications. *Matern Health Neonatol Perinatol* 3, 28, doi:10.1186/s40748-017-0068-7 (2017). [PubMed: 29234502]
32. Yao A, Moinian M & Lind J Distribution of blood between infant and placenta after birth. *The Lancet* 294, 871–873 (1969).
33. Naulaers G, Barten S, Vanhole C, Verhaeghe J & Devlieger H Management of severe neonatal anemia due to fetomaternal transfusion. *Am J Perinatol* 16, 193–196, doi:10.1055/s-2007-993857 (1999). [PubMed: 10458533]
34. Polglase GR, Ong T & Hillman NH Cardiovascular Alterations and Multiorgan Dysfunction After Birth Asphyxia. *Clin Perinatol* 43, 469–483, doi:10.1016/j.clp.2016.04.006 (2016). [PubMed: 27524448]
35. ACOG Practice Bulletin No. 106: Intrapartum fetal heart rate monitoring: nomenclature, interpretation, and general management principles. *Obstet Gynecol* 114, 192–202, doi:10.1097/AOG.0b013e3181aef106 (2009). [PubMed: 19546798]
36. Tao E, Ye D, Long G, Hu Y, Fu Q, Yuan T et al. Severe neonatal anemia affected by massive fetomaternal hemorrhage: a single-center retrospective observational study. *The Journal of Maternal-Fetal & Neonatal Medicine* 35, 3972–3978, doi:10.1080/14767058.2020.1845313 (2022). [PubMed: 33183095]
37. Sá I & Moura M Sinusoidal pattern: a key to a rare case of fetal anaemia. *BMJ Case Rep* 14, e241312, doi:10.1136/bcr-2020-241312 (2021).
38. Mayer C, Rijhsinghani A & Parikh R The Limitations of Middle Cerebral Artery Peak Systolic Velocity and the Need for Certification [38A]. *Obstetrics & Gynecology* 135 (2020).
39. Sueters M, Arabin B & Oepkes D Doppler sonography for predicting fetal anemia caused by massive fetomaternal hemorrhage. *Ultrasound in Obstetrics & Gynecology* 22, 186–189, doi:10.1002/uog.190 (2003). [PubMed: 12905516]

40. Weiner GM, Zaichkin J & American Academy of Pediatrics, A. H. A. Textbook of Neonatal Resuscitation, 8th Edition. (2021).
41. Savithrisowmya S, Singh M, Kriplani A, Agarwal N, Mehra NK & Bhatla N Assessment of fetomaternal hemorrhage by flow cytometry and Kleihauer-Betke test in Rh-negative pregnancies. *Gynecol Obstet Invest* 65, 84–88, doi:10.1159/000108401 (2008). [PubMed: 17878734]
42. Kim YA & Makar RS Detection of fetomaternal hemorrhage. *American Journal of Hematology* 87, 417–423, doi:10.1002/ajh.22255 (2012). [PubMed: 22231030]
43. Mercer JS, Erickson-Owens DA & Rabe H Placental transfusion: may the “force” be with the baby. *J Perinatol* 41, 1495–1504, doi:10.1038/s41372-021-01055-0 (2021). [PubMed: 33850284]
44. Aziz K, Lee HC, Escobedo MB, Hoover AV, Kamath-Rayne BD, Kapadia VS et al. Part 5: Neonatal Resuscitation: 2020 American Heart Association Guidelines for Cardiopulmonary Resuscitation and Emergency Cardiovascular Care. *Circulation* 142, S524–S550, doi:10.1161/CIR.0000000000000902 (2020). [PubMed: 33081528]
45. Shalish W, Olivier F, Aly H & Sant’Anna G in *Seminars in Fetal and Neonatal Medicine*. 328–335 (Elsevier).
46. Todd SR, Malinoski D, Muller PJ & Schreiber MA Lactated Ringer’s is Superior to Normal Saline in the Resuscitation of Uncontrolled Hemorrhagic Shock. *Journal of Trauma and Acute Care Surgery* 62 (2007).
47. Healey MA, Davis RE, Liu FC, Loomis WH & Hoyt DB Lactated Ringer’s is superior to normal saline in a model of massive hemorrhage and resuscitation. *Journal of Trauma and Acute Care Surgery* 45, 894–899 (1998).
48. McIntyre L, Fergusson D, McArdle T, English S, Cook DJ, Fox-Robichaud AE et al. A Crossover Trial of Hospital-Wide Lactated Ringer’s Solution versus Normal Saline. *New England Journal of Medicine* (2025).
49. ACOG Practice Bulletin No. 192: Management of Alloimmunization During Pregnancy. *Obstet Gynecol* 131, e82–e90, doi:10.1097/aog.0000000000002528 (2018). [PubMed: 29470342]
50. Stephenson T, Zuccollo J & Mohajer M Diagnosis and management of non-immune hydrops in the newborn. *Arch Dis Child Fetal Neonatal Ed* 70, F151–154, doi:10.1136/fn.70.2.f151 (1994). [PubMed: 8154908]
51. Katheria AC, Rich WD & Lakshminrusimha S Circulatory emergencies in the delivery room. *Semin Fetal Neonatal Med* 24, 101030, doi:10.1016/j.siny.2019.101030 (2019). [PubMed: 31563413]
52. Rubod C, Deruelle P, Le Goueff F, Tunez V, Fournier M & Subtil D Long-Term Prognosis for Infants After Massive Fetomaternal Hemorrhage. *Obstetrics & Gynecology* 110 (2007).

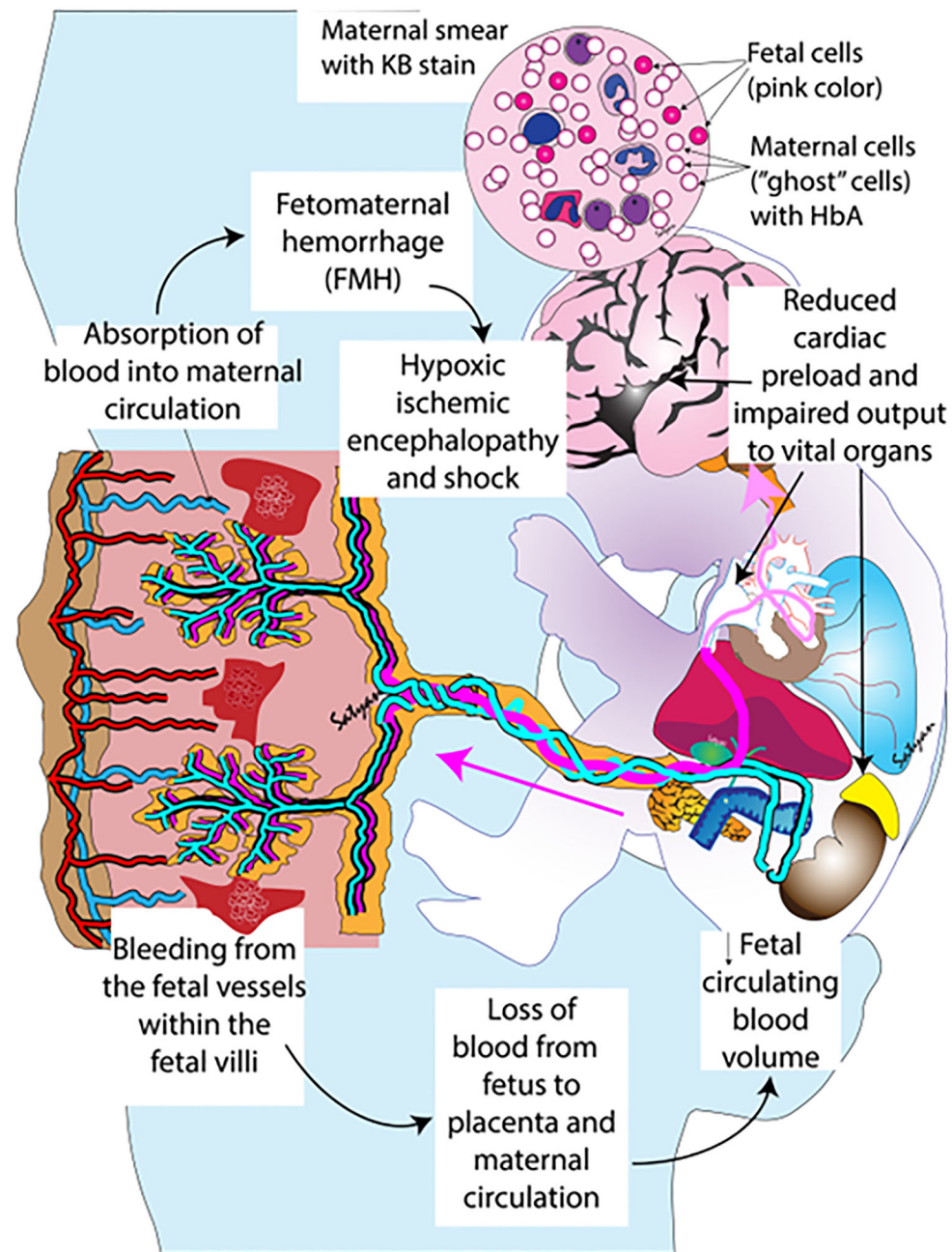
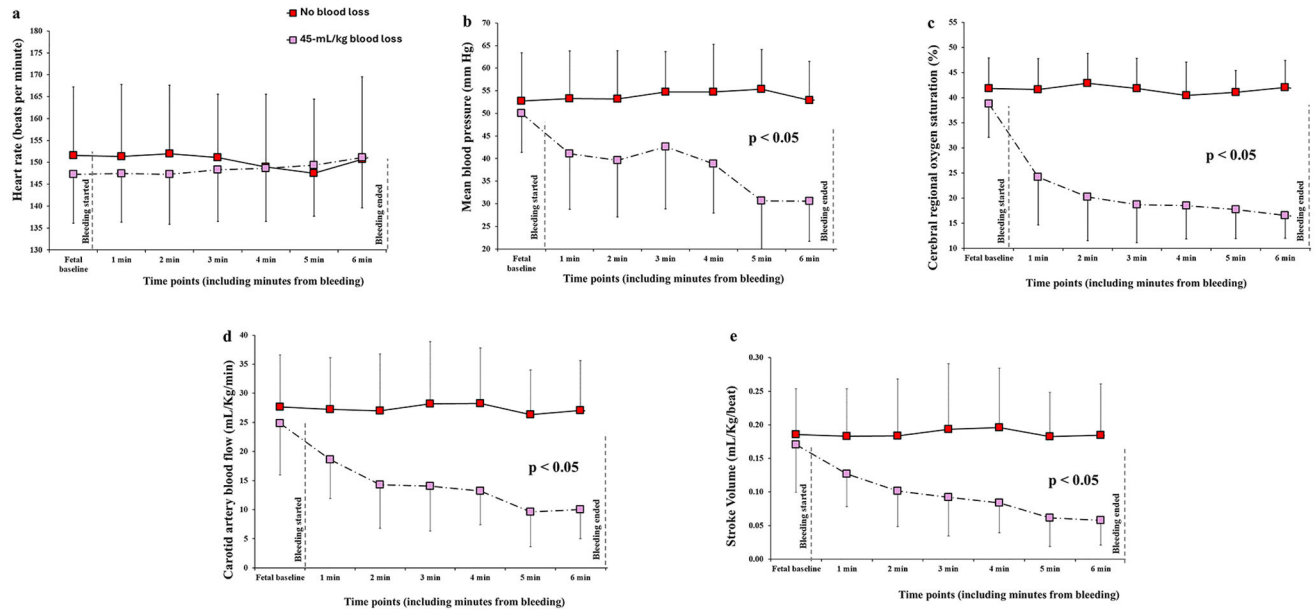


Figure 1:

Pathophysiology of acute fetomaternal hemorrhage (FMH). Bleeding from the fetal blood vessels within the fetal aspect of the chorionic villi is followed by the absorption of fetal blood into the maternal circulation. This results in reduced cardiac preload, impaired output into vital organs and decreased fetal circulating blood volume. Consequences of FMH on the neonate include hypoxic ischemic encephalopathy and hypovolemic shock. Kleihauer Betke test (KB stain) reveals pink-stained fetal cells due to presence of fetal hemoglobin (HbF) surrounded by “ghost” maternal red blood cells containing hemoglobin A (HbA). Copyright Satyan Lakshminrusimha- used with permission.

**Figure 2:**

Change in hemodynamics during blood loss from fetomaternal circulation. Hemodynamics were compared between lambs without blood loss prior to umbilical cord occlusion and lambs with 45-mL/kg blood loss prior to umbilical cord occlusion. The x axis represents time points between the start of exsanguination (removal of 45 mL/kg of blood) and final time point of umbilical cord occlusion. There was no change in fetal heart rate during the period of acute blood loss of 45 mL/kg from the fetomaternal circulation (2a). However, there was a significant decrease in mean blood pressure (2b), a cerebral regional oxygen saturation (2c), left carotid artery blood flow (2d), and stroke volume (2e) during removal of 45-mL/kg from the fetoplacental circulation.

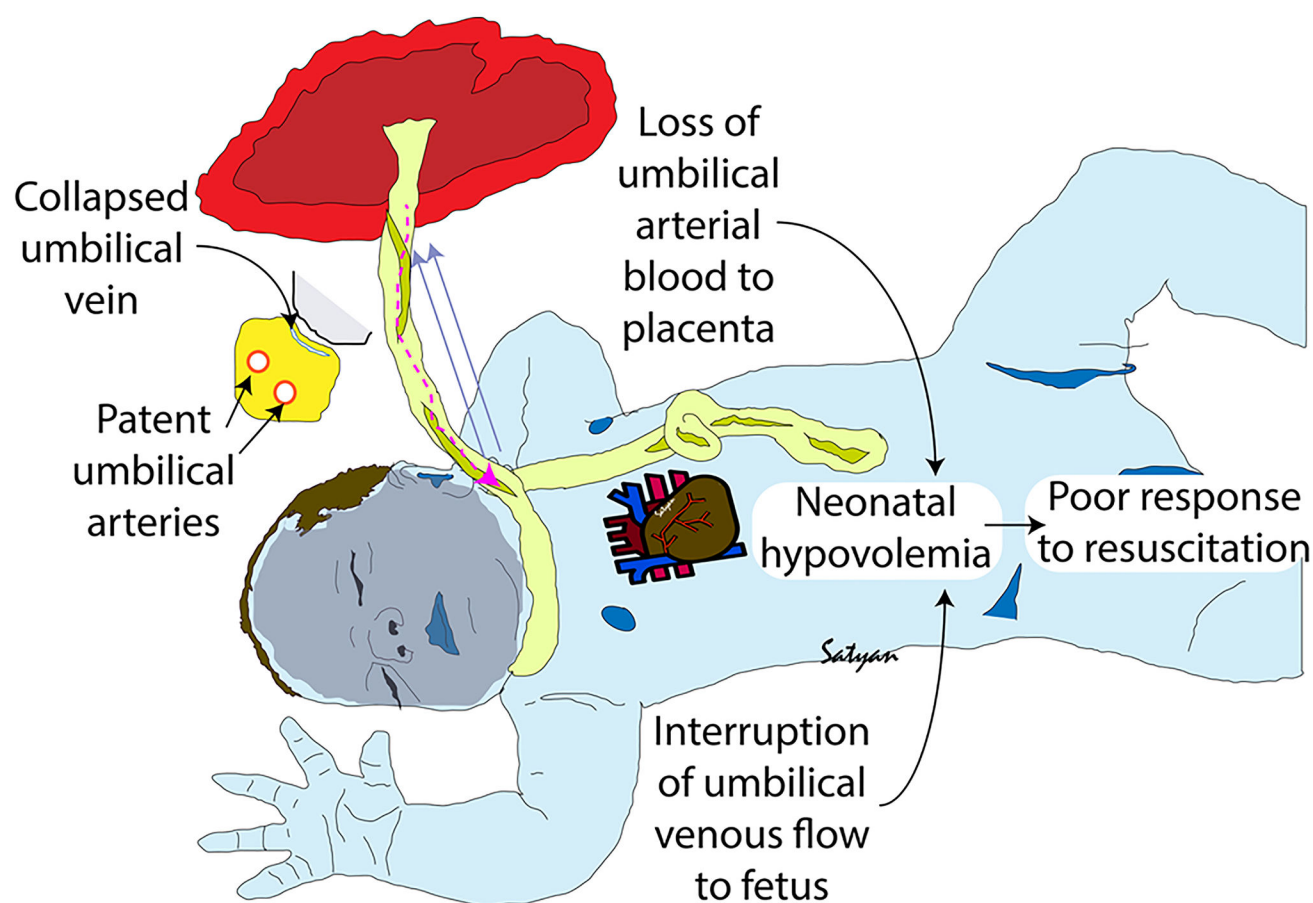


Figure 3: Fetoplacental hemorrhage in cases of tight nuchal cord. A newborn delivered through a tight nuchal cord have collapsed thin-walled umbilical vein and patent thick-walled umbilical artery, resulting in neonatal hypovolemia impairing response to resuscitation. Copyright Satyan Lakshminrusimha- used with permission.

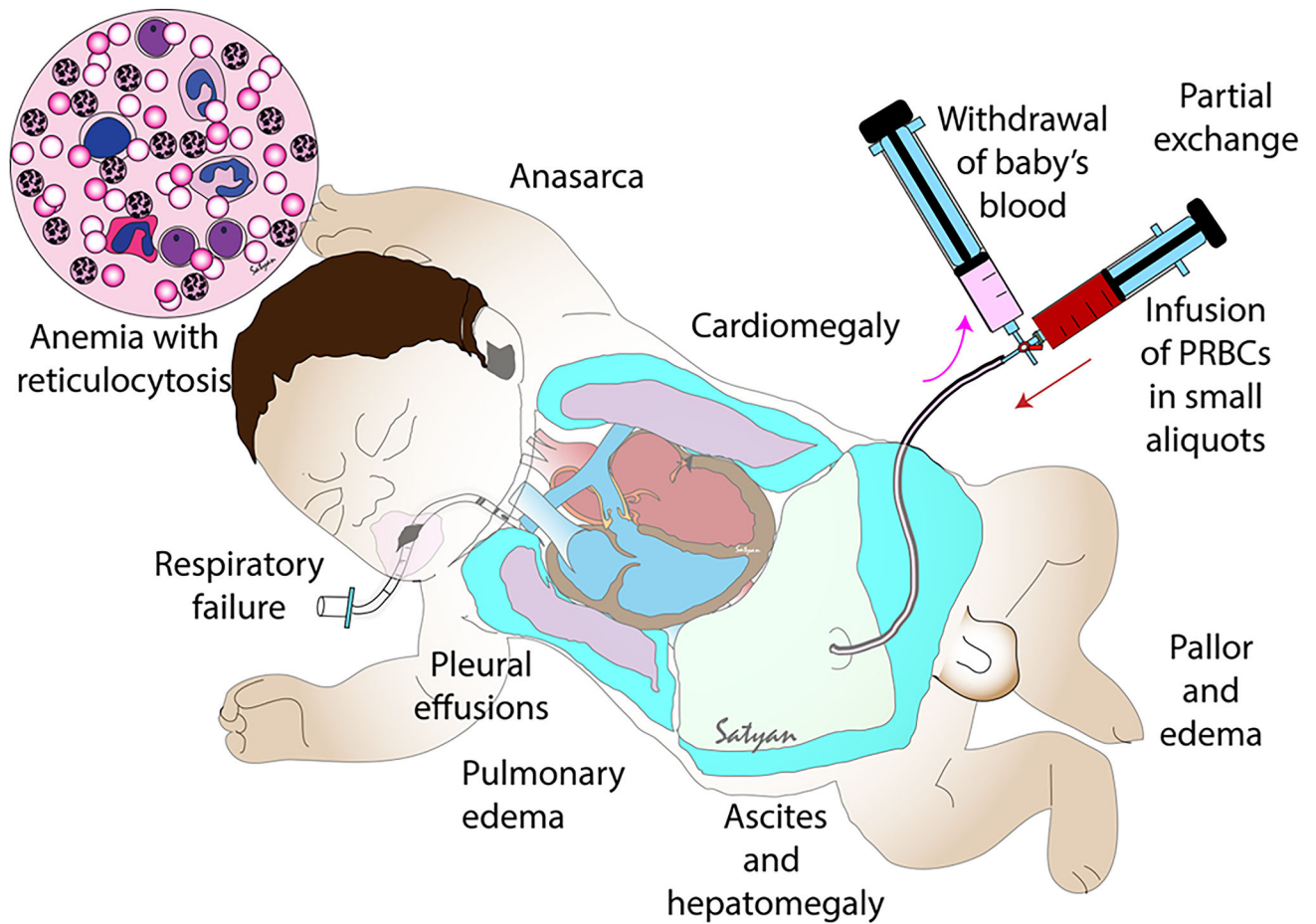


Figure 4:

Clinical features in neonates affected by chronic fetomaternal hemorrhage (FMH).

Newborns affected by clinically significant FMH can present with respiratory failure due to pulmonary edema and pleural effusion, pallor, edema, ascites, hepatomegaly and cardiomegaly. Partial exchange transfusion may be required to manage chronic neonatal anemia, with withdrawal of baby's blood and infusion of PRBCs in small aliquots. The inset shows peripheral smear finding of anemia with reticulocytosis. Copyright Satyan Lakshminrusimha- used with permission.

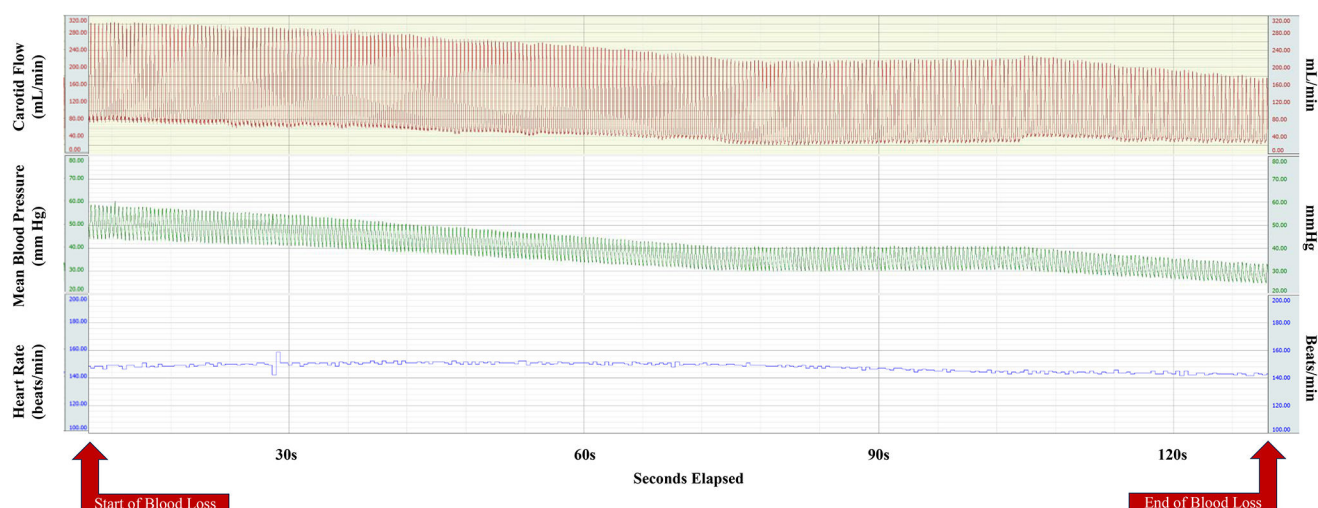


Figure 5:
Biopac snapshot during a 2-minute period of fetal blood loss. The biopac snapshot demonstrates lack of change in heart rate and decrease in carotid artery blood flow and mean blood pressure during fetal blood loss of ~20 ml/kg over 2 minutes from a fetal lamb in placental circulation. Copyright Deepika Sankaran.

Table 1:
Maternal obstetrical conditions associated with fetomaternal hemorrhage.

Cause	Incidence	Etiopathogenesis	Risk Factors
Placental abruption ¹⁸	0.6 – 1.2% of pregnancies	Premature separation of the placenta from uterine wall accompanied by rupture of maternal blood vessels, bleeding into decidua-placental interface with a potential for fetal blood loss into maternal circulation.	Prematurity, maternal age, parity, tobacco use, maternal hypertension, history of cesarean section. 10-fold increased risk of recurrence in 2 nd pregnancy and 25-fold increased risk of recurrence in 3 rd pregnancy after initial pregnancy with abruption.
Placenta previa ¹⁹	0.4–0.5% of births	When the placenta lays low in the uterus, partly or completely covering the cervical os that leads to shearing of placental vessels with lower uterine segment changes, or during cervical dilation resulting in fetomaternal hemorrhage (FMH). This bleeding is painless.	History of cesarean section, multiple gestation
Vasa previa ²⁰	0.08–0.04% of pregnancies (around 1 in 2500 deliveries)	Fetal (unprotected) vessels traverse membranes near the cervical os (often associated with velamentous cord insertion) that rupture leading to direct FMH or fetal exsanguination.	Assisted reproductive techniques, low-lying placenta, pregnancy with multiple gestation.
Trauma (abdominal, motor vehicle accident, physical assault in pregnancy) ²¹	Less than half of maternal traumas result in FMH. Unintentional trauma during pregnancies: 207 per 100,000 pregnancies.	Mechanical disruption of placental interface or fetal vessels leading to leakage of fetal blood into maternal circulation.	No identifiable risk factors; gestational age at time of trauma and type of trauma are not correlated to FMH.
Placenta accreta spectrum ²²	1 in 272 women who had a birth-related hospital discharge	Defect of the endometrial–myometrial interface followed by failure of normal decidualization in the area of a uterine scar, which allows abnormally deep placental anchoring villi and trophoblast infiltration.	0.3% in women with one previous cesarean delivery to 6.74% for women with five or more cesarean deliveries. Additional risk factors include advanced maternal age, multiparity, prior uterine surgeries or curettage, placenta previa and Asherman syndrome.
Umbilical Cord Prolapse ²³	0.11–0.18% of births	The cord descends past or alongside the presenting fetal part (typically after membrane rupture) followed by compromise of fetal circulation and potentially umbilical vessel injury.	Cervical dilation and station of presenting part at time of rupture of membranes
Umbilical Cord Avulsion ²⁴	~ 0.08% of births	Avulsion (cord “snapping” or vessels rupturing) leads to direct fetal blood loss into maternal side or external hemorrhage	Abnormal cord insertion, abnormal cord length/size
Tight Nuchal Cord ²⁵	3.8–6.6% of births	The structure of the umbilical cord is such that twisting and knotting may lead to obstruction of the collapsible thin-walled umbilical vein, but the thicker walled umbilical arteries remain patent resulting in hypovolemia, acidosis and anemia.	Advancing gestational age, length of labor

Table 2.

Review of gaps in current knowledge regarding diagnosis and management of fetomaternal hemorrhage (FMH)

Gap in knowledge	Description
Timely recognition of FMH and fetal blood loss	What is the earliest sign of FMH utilizing currently available tools for maternal-fetal monitoring? Is fetal heart rate a reliable and early indicator of fetal compromise from FMH? Does monitoring of fetal oxygenation status help in early recognition of fetal blood loss? How can we identify occult blood loss? Is there a role for testing for markers such as plasma arginine vasopressin that may increase in the fetus due to plasma volume contraction?
Indication for volume expansion during neonatal resuscitation	How to distinguish between euvolemic and hypovolemic asphyxia? Is administration of fluid bolus harmful in euvolemic asphyxia, or in the absence of blood loss?
Optimal timing of volume expander during neonatal resuscitation	Is there a role for early volume expansion with administration of fluid bolus prior to first dose of intravenous epinephrine in an infant not responding to ventilation and chest compressions? After lack of response to how many doses of intravenous epinephrine should a volume expander be considered?
Speed of volume expansion	Should the volume expander be administered over 5–10 minutes or faster (<2 minutes)? Should the volume expander be used as a large flush volume after the intravenous epinephrine dose?
Choice of volume expander	What is the ideal volume expander in the delivery room? Crystalloids (normal saline or lactated Ringer's solution), colloids such as 5% albumin, or emergency release blood products (whole blood vs packed red blood cells)? Is there a role for acetate or gluconate buffered colloids such as sodium acetate during neonatal resuscitation? Is it possible to replenish fetal hemoglobin with high oxygen affinity that was lost from the fetus during FMH? Is there a role for femoral occlusion during resuscitation after acute fetal blood loss? Does umbilical cord milking help in improving circulating blood volume after FMH?
Role of alternate medications	Is there a role for vasopressin to increase systemic vascular resistance or calcium to increase cardiac contractility?